

API Endpoint Plan

2. Authentication

Method	Route	Description	Role required	Request body	Expected response
POST	/api/auth/register	Creates a new account and, when the role is Participant, the linked participant profile. Passwords are hashed with BCrypt before storage.	None	{ email, password, confirmPassword, firstName, lastName, phoneNumber, role, dateOfBirth, gender, clubName, tShirtSize, emergencyContactName, emergencyContactPhone }	201 Created — { userId, email, role } 400 Bad Request — validation failed (weak password, passwords do not match, missing profile fields for a Participant) 409 Conflict — email already registered
POST	/api/auth/login	Verifies credentials and issues a JWT containing the user id and role claim.	None	{ email, password }	200 OK — { token, expiresAt, userId, fullName, role } 400 Bad Request — missing fields 401 Unauthorized — invalid credentials or deactivated account
POST	/api/auth/change-password	Lets a signed-in user change their own password after confirming the current one.	Any	{ currentPassword, newPassword, confirmNewPassword }	204 No Content — password updated 400 Bad Request — new passwords do not match or fail policy 401 Unauthorized — current password incorrect
GET	/api/auth/roles	Returns the role list so the registration form can bind a dropdown instead of hard-coding strings.	None	None	200 OK — [{ roleId, roleName, description }]

3. User profile

Method	Route	Description	Role required	Request body	Expected response
GET	/api/users/me	Returns the signed-in user's account details, plus the participant profile when the role is Participant.	Any	None	200 OK — { userId, email, firstName, lastName, phoneNumber, role, createdAt, profile } 404 Not Found — user no longer exists
PUT	/api/users/me	Updates the signed-in user's own account details.	Any	{ firstName, lastName, phoneNumber }	200 OK — updated user 400 Bad Request — validation failed

Method	Route	Description	Role required	Request body	Expected response
PUT	/api/users/me/profile	Creates or updates the participant-only profile (club, shirt size, emergency contact, medical notes).	Participant	{ dateOfBirth, gender, clubName, tShirtSize, emergencyContactName, emergencyContactPhone, medicalNotes }	200 OK — updated profile 400 Bad Request — date of birth in the future or invalid gender/size 403 Forbidden — caller is an Organiser
GET	/api/users/{id}	Lets an Organiser view a participant's contact and emergency details for an event they organise (race-day safety requirement).	Organiser	None	200 OK — { userId, fullName, email, phoneNumber, profile } 403 Forbidden — the user is not entered in any of the caller's events 404 Not Found — user does not exist
DELETE	/api/users/me	Soft-deletes the signed-in account by setting IsActive = 0; historical results are retained.	Any	None	204 No Content — account deactivated 409 Conflict — Organiser still has published events with confirmed entries

4. Events

Method	Route	Description	Role required	Request body	Expected response
GET	/api/events	Lists published events for the public browse page. Supports ?search=, ?province=, ?eventType=, ?fromDate=, ?toDate=, ?page=, ?pageSize=.	None	None	200 OK — paged list of { eventId, name, eventType, eventDate, city, province, status, categoryCount, bannerUri }
GET	/api/events/{id}	Returns one event with its categories, organiser name and remaining capacity per category.	None	None	200 OK — full event detail 404 Not Found — event does not exist or is a draft belonging to another organiser
POST	/api/events	Creates a new event owned by the calling Organiser.	Organiser	{ name, description, eventType, eventDate, venueName, city, province, latitude, longitude, registrationOpensOn, registrationClosesOn, status }	201 Created — created event plus Location header 400 Bad Request — closing date after the event date, unknown province, or event date in the past 403 Forbidden — caller is a Participant
PUT	/api/events/{id}	Updates an event the caller owns.	Organiser	Same fields as POST	200 OK — updated event 400 Bad Request — validation failed 403 Forbidden — caller does not own the event 404 Not Found — event does not exist

Method	Route	Description	Role required	Request body	Expected response
DELETE	/api/events/{id}	Deletes an event that has no confirmed entries; otherwise the organiser must cancel it instead.	Organiser	None	204 No Content — deleted 403 Forbidden — not the owner 404 Not Found — event does not exist 409 Conflict — event has confirmed enrolments
PATCH	/api/events/{id}/status	Moves an event between Draft, Published, Cancelled and Completed without touching the rest of the record.	Organiser	{ status }	200 OK — { eventId, status } 400 Bad Request — invalid transition (e.g. Completed back to Draft) 403 Forbidden — not the owner
GET	/api/events/mine	Lists every event created by the calling Organiser, including drafts, with entry counts.	Organiser	None	200 OK — list of { eventId, name, eventDate, status, entryCount, revenue }
GET	/api/events/{id}/enrolments	Returns the full entry list for one of the caller's events — the race-day start list. Supports ?categoryId= and ?status=.	Organiser	None	200 OK — [{ enrolmentId, raceNumber, participantName, categoryName, status, amountPaid, hasResult }] 403 Forbidden — not the owner 404 Not Found — event does not exist
GET	/api/events/{id}/weather	Proxies a third-party weather API using the event's latitude and longitude so participants can prepare for race day.	None	None	200 OK — { eventDate, temperatureC, feelsLikeC, conditions, windKph, chanceOfRain } 404 Not Found — event does not exist 503 Service Unavailable — weather provider unreachable

5. Event categories

Method	Route	Description	Role required	Request body	Expected response
GET	/api/events/{eventId}/categories	Lists the distance options for an event, with entry fee, start time and places remaining.	None	None	200 OK — [{ categoryId, name, distanceKm, entryFee, startTime, maxParticipants, placesRemaining, minimumAge }] 404 Not Found — event does not exist
POST	/api/events/{eventId}/categories	Adds a category to an event the caller owns.	Organiser	{ name, distanceKm, entryFee, startTime, maxParticipants, minimumAge, cutOffMinutes }	201 Created — created category 400 Bad Request — distance or fee not positive 403 Forbidden — not the owner 409 Conflict — a category with that name already exists on the event

Method	Route	Description	Role required	Request body	Expected response
GET	/api/categories/{id}	Returns a single category with its parent event summary.	None	None	200 OK — category detail 404 Not Found — category does not exist
PUT	/api/categories/{id}	Updates a category the caller owns.	Organiser	Same fields as POST	200 OK — updated category 400 Bad Request — maxParticipants lower than the number already entered 403 Forbidden — not the owner 404 Not Found — category does not exist
DELETE	/api/categories/{id}	Removes a category that nobody has entered yet.	Organiser	None	204 No Content — deleted 403 Forbidden — not the owner 409 Conflict — category has enrolments

6. Route waypoints

Method	Route	Description	Role required	Request body	Expected response
GET	/api/categories/{categoryId}/waypoints	Returns the ordered route points (start, water points, markers, finish) used to draw the route map.	None	None	200 OK — [{ waypointId, sequenceNumber, name, waypointType, distanceMarkerKm, latitude, longitude }] 404 Not Found — category does not exist
POST	/api/categories/{categoryId}/waypoints	Adds a waypoint to a category's route.	Organiser	{ sequenceNumber, name, waypointType, distanceMarkerKm, latitude, longitude }	201 Created — created waypoint 400 Bad Request — latitude or longitude out of range 403 Forbidden — not the owner 409 Conflict — that sequence number is already used on the route
PUT	/api/waypoints/{id}	Updates a single waypoint.	Organiser	Same fields as POST	200 OK — updated waypoint 403 Forbidden — not the owner 404 Not Found — waypoint does not exist
DELETE	/api/waypoints/{id}	Removes a waypoint from the route.	Organiser	None	204 No Content — deleted 403 Forbidden — not the owner 404 Not Found — waypoint does not exist

7. Event enrolments

Method	Route	Description	Role required	Request body	Expected response
POST	/api/events/{eventId}/enrolments	Enters the signed-in Participant into one category of an event and allocates a race number.	Participant	{ categoryId, amountPaid, paymentReference }	201 Created — { enrolmentId, raceNumber, categoryName, status, amountPaid } 400 Bad Request — the category does not belong to this event, or the participant is below the minimum age 403 Forbidden — caller is an Organiser 404 Not Found — event or category does not exist 409 Conflict — already entered for this event, registration closed, or the category is full
GET	/api/enrolments/me	Returns the signed-in Participant's own entries, upcoming and past. Supports ?status= and ?upcoming=true.	Participant	None	200 OK — [{ enrolmentId, eventName, eventDate, categoryName, distanceKm, raceNumber, status, hasResult }]
GET	/api/enrolments/{id}	Returns one entry in full — the digital race entry confirmation.	Any	None	200 OK — enrolment detail with event, category and result 403 Forbidden — not the participant who entered, nor the organiser of that event 404 Not Found — enrolment does not exist
PUT	/api/enrolments/{id}/category	Lets a Participant switch distance before registration closes (e.g. 42.2 km down to 21.1 km).	Participant	{ categoryId }	200 OK — updated enrolment 400 Bad Request — target category is in a different event 403 Forbidden — not the owner of the entry 409 Conflict — registration has closed or the target category is full
PATCH	/api/enrolments/{id}/status	Lets an Organiser confirm or cancel an entry, for example after verifying payment.	Organiser	{ status }	200 OK — { enrolmentId, status } 403 Forbidden — not the organiser of that event 404 Not Found — enrolment does not exist
DELETE	/api/enrolments/{id}	Withdraws the signed-in Participant from an event by setting the status to Withdrawn; the record is retained for the organiser's records.	Participant	None	204 No Content — withdrawn 403 Forbidden — not the owner of the entry 409 Conflict — the event has already taken place

8. Results

Method	Route	Description	Role required	Request body	Expected response
POST	/api/enrolments/{enrolmentId}/result	Captures the finishing time and positions for one entry after the race.	Organiser	{ finishTimeSeconds, overallPosition, categoryPosition, status }	201 Created — created result 400 Bad Request — a Finished result without a time, or a negative time 403 Forbidden — not the organiser of that event 404 Not Found — enrolment does not exist 409 Conflict — a result already exists for this entry
PUT	/api/results/{id}	Corrects a previously captured result.	Organiser	Same fields as POST	200 OK — updated result 400 Bad Request — validation failed 403 Forbidden — not the organiser of that event 404 Not Found — result does not exist
DELETE	/api/results/{id}	Removes a result that was captured against the wrong race number.	Organiser	None	204 No Content — deleted 403 Forbidden — not the organiser of that event 404 Not Found — result does not exist
GET	/api/events/{eventId}/results	Returns the public results board for an event, ordered by category and position. Supports ?categoryId=.	None	None	200 OK — [{ raceNumber, participantName, categoryName, finishTimeSeconds, overallPosition, categoryPosition, status }] 404 Not Found — event does not exist
GET	/api/results/me	Returns the signed-in Participant's full race history for their performance page.	Participant	None	200 OK — [{ resultId, eventName, eventDate, categoryName, distanceKm, finishTimeSeconds, paceSecondsPerKm, categoryPosition, status }]
GET	/api/results/me/stats	Returns aggregated personal statistics: events completed, total distance, best time per distance and average pace.	Participant	None	200 OK — { eventsEntered, eventsCompleted, totalDistanceKm, personalBests: [{ distanceKm, finishTimeSeconds, eventName }], averagePaceSecondsPerKm }
POST	/api/events/{eventId}/results/bulk	Uploads a whole field of results in one call from the timing system export.	Organiser	[{ raceNumber, finishTimeSeconds, overallPosition, categoryPosition, status }]	200 OK — { imported, skipped, errors: [{ raceNumber, reason }] } 403 Forbidden — not the organiser of that event 404 Not Found — event does not exist

9. Event media (part 3 planning)

Method	Route	Description	Role required	Request body	Expected response
GET	/api/events/{eventId}/media	Lists the banner, route map, gallery images and documents attached to an event.	None	None	200 OK — [{ mediaId, mediaType, fileName, blobUri, contentType, sizeInBytes, uploadedAt }] 404 Not Found — event does not exist

Method	Route	Description	Role required	Request body	Expected response
POST	/api/events/{eventId}/media	Uploads a file to the Azure Blob container and stores its URI against the event. Sent as multipart/form-data.	Organiser	file (binary) + mediaType	201 Created — { mediaId, blobUri } 400 Bad Request — unsupported content type or file larger than 5 MB 403 Forbidden — not the owner 404 Not Found — event does not exist
DELETE	/api/media/{id}	Deletes the blob and its metadata row.	Organiser	None	204 No Content — deleted 403 Forbidden — not the owner 404 Not Found — media does not exist

10. System

Method	Route	Description	Role required	Request body	Expected response
GET	/api/health	Liveness probe that also confirms the database connection. Used by the CI/CD pipeline and the Docker health check in Part 3.	None	None	200 OK — { status: "Healthy", database: "Connected", version } 503 Service Unavailable — database unreachable